EMMANUEL M NDALIRO

Software Engineer — Full Stack (React / Python / AI)

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PROFILE

Full-stack software engineer with 6+ years of experience building and shipping web applications end-to-end — React/Next.js front ends, Python (Flask/Django) backends, and cloud deployment. I own features from idea to production: recent example is a production AI-powered job-search platform I designed, built and deployed solo (Next.js 14 + TypeScript + Tailwind dashboard over a Flask API, with browser automation and LLM services, live on Render behind a full CI/CD pipeline).

What sets me apart for a product like Swavy's: I don't just call LLM APIs — I build the agent systems around them (orchestration, approval gates, validation pipelines), and I ship the complete product around them. Strong written communicator (published technical author) and comfortable working remotely across time zones.

CORE STACK

- **Frontend:** React, Vue.js, Next.js 14 (App Router), TypeScript, JavaScript (ES6+), Tailwind CSS, HTML5, CSS, React Server/Client Components, REST integration, responsive/mobile-first UI, WebSockets
- **Backend:** Python (Flask, Django, FastAPI), RESTful API design, WebSocket, authentication, background jobs
- **AI Engineering:** LLM integration & orchestration (OpenAI/Anthropic/OpenRouter-compatible clients), RAG, agentic workflows with human-approval gates, prompt engineering, output validation pipelines
- **Cloud & DevOps:** AWS/Azure/GCP, Docker, CI/CD (GitHub Actions), Render deployment, monitoring & logging
- **Automation & Testing:** Playwright, Selenium, pytest; web scraping (BeautifulSoup, AI-assisted extraction)
- **Also:** Flutter/Dart (cross-platform mobile), Firebase, PostgreSQL/SQLite, Git/GitHub

KEY PROJECTS

AI Job-Search Platform — solo full-stack build (2026)

Next.js 14 · TypeScript · Tailwind CSS · Flask · Playwright · Claude/OpenRouter · Render CI/CD

- Built and deployed a complete production platform: a **Next.js 14 + TypeScript + Tailwind** dashboard (14+ pages: job feed, matching insights, AI copilot chat, application tracker, interview scheduler, analytics) over a **Flask REST API** with 59 endpoints.
- Implemented Playwright-based browser automation that fills job applications across ATS platforms (Greenhouse, Lever, Ashby), with an LLM field-mapping layer and human-approval gates before anything submits.
- Shipped a full CI/CD pipeline on GitHub Actions (tests, builds, deploy hooks, post-deploy smoke tests) with zero-downtime renders on Render.
- Wrote an agent architecture that scores, explains and tracks job matches — the same "agents with approval checkpoints" pattern used by modern AI products.

Agentic AI Systems & Published Technical Author (2025 — present)

Python · LangGraph · Pydantic AI · MCP/A2A protocols

- Designed structured multi-agent systems and documented them publicly: "The AI Protocol Wars: MCP vs. A2A — A Comprehensive Analysis with Practical Implementations" and "Building Structured AI Agents" (Medium + open-source code on GitHub).
- Built context-engineering strategies and validation pipelines that make LLM output production-reliable rather than demo-reliable.

LLM Applications & Cloud Integration (2023 — 2025)

Python · Flask/Django · Hugging Face · Ollama · AWS/Azure/GCP

- Built and deployed LLM-powered applications: RAG, Agentic RAG, Hybrid RAG, and LLM chaining for real-world language tasks — with validation pipelines using open-source datasets to score and improve response quality.
- Integrated AI services with secure Python backends and cloud infrastructure (AWS/Azure/GCP), including serverless inference via Lambda/cloud functions; automated data flows, QA and business processes in Agile teams.
- Architected containerized microservices with CI/CD pipelines for automated testing and deployment; implemented monitoring and logging for proactive issue detection.

Cross-Platform Mobile & Web Automation (2021 — 2022)

Flutter/Dart · Firebase · Python automation

- Built Flutter applications consuming AI-powered APIs; automated workflows with UIPath, Selenium and Playwright; managed Flask/Django backends with Firebase/SQLite and RESTful integrations, including location-based features on Google Maps APIs.
- Certified in React (React for Web Designers) and Vue.js (Vue.js Essential Training) — formal front-end credentials backing the JavaScript work across these projects.

EDUCATION & CERTIFICATIONS

Education

- **BSc. Software Engineering** — Multimedia University of Kenya (2015–2019), 2nd Class Honours

Frontend & Web Certifications *(LinkedIn Learning)*

- React for Web Designers
- Vue.js Essential Training
- Learning Google Firebase for Flutter

Engineering & Cloud Certifications

- Google Africa Developer Training Program — Advanced Cloud & Development Track *(Google Developers)*
- Android Development Associate — Google Africa Developer Scholarship 2020 *(Google Developers)*
- Spring: Test-Driven Development with JUnit *(LinkedIn Learning)*
- Introduction to Data Science *(LinkedIn Learning)*

Quality & Automation Certifications *(Test Automation University)*

- Foundation for Test Automation (Credential ID: 1531e308) · Python Programming · Source Control for Test Automation · Automated Testing

WHY SWAVY (short version)

Swavy's product is AI agents that run the influencer-marketing pipeline with human approval gates — that is precisely the architecture I build: agentic backend + polished React front end + human-in-the-loop control. I want to help build what's next, and I'm used to owning it end-to-end.

References and writing samples available on request.